

Test of Mathematics for University Admission
Mock Set E Paper 2
 Mathematical Reasoning

Name: _____

Date sat: _____

Score: _____ / 20

Answer all 20 questions. Each question has exactly one correct option and is worth one mark. No marks are deducted for wrong answers. Throughout, “must be true” means true in every case allowed by the information given, and a counterexample must satisfy the hypothesis while failing the conclusion.

[illegible]

Question 1

The statement “if a quadrilateral is a rhombus, then its diagonals are perpendicular” is true. Which of the following is a counterexample to its *converse*?

- A a square
- B a rectangle that is not a square
- C a kite that is not a rhombus
- D a parallelogram that is not a rhombus
- E a rhombus that is not a square
- F there is none: the converse is also true

Question 2

Which of the following statements about real numbers are true?

- I For every x there is a y with $y > x$.
 - II There is a y such that $y > x$ for every x .
 - III For every x there is a y with $y^2 = x$.
 - IV There is an x such that $xy = x$ for every y .
- A I only
 - B I and IV only
 - C I, II and IV only
 - D IV only
 - E I and III only
 - F I, III and IV only

Question 3

For a quadratic function f , the condition “ $f(1)$ and $f(3)$ have opposite signs” is _____ for “ $f(x) = 0$ has at least one root strictly between 1 and 3”.

- A necessary but not sufficient
- B sufficient but not necessary
- C necessary and sufficient
- D neither necessary nor sufficient

Question 4

Call a positive integer *neat* if it can be written as the difference of the squares of two integers. Which of the following statements are true?

- I Every odd positive integer is neat.
 - II The product of two neat numbers is neat.
 - III The sum of two neat numbers is neat.
 - IV If n is neat, then $2n$ is neat.
- A I only
 - B II only
 - C I and II only
 - D I, II and III only
 - E I and IV only
 - F I, II, III and IV

Question 5

A student solves

$$\frac{x^2 - 4}{x - 2} = x^2 - 2x + 4$$

as follows: “The left-hand side simplifies to $x + 2$, so $x + 2 = x^2 - 2x + 4$, giving $x^2 - 3x + 2 = 0$ and $x = 1$ or $x = 2$.” Which of the following is correct?

- A The working and the answer are both correct.
- B The simplification is invalid, and in fact there are no solutions.
- C The quadratic was solved incorrectly.
- D $x = 2$ must be rejected, so the only solution is $x = 1$.
- E $x = 1$ must be rejected, so the only solution is $x = 2$.
- F There are three solutions: 1, 2 and -2 .

Question 6

What is the largest positive integer d with the property that, for every integer n , if n^2 is divisible by 12 then n is divisible by d ?

- A 2
- B 3
- C 4
- D 6
- E 12
- F 24

Question 7

A function f satisfies $f'(x) = (x - 1)^2(x - 3)$ for all x . Which of the following must be true?

- I f has exactly one local minimum.
 - II $f(0) > f(2)$
 - III f has a local maximum at $x = 1$.
 - IV $f(4) > f(3)$
- A I only
 - B I and IV only
 - C I, II and IV only
 - D I, II, III and IV
 - E II and III only
 - F I and III only

Question 8

A student proves: *for all real x and y , if $x > y$ then $x^2 > y^2$.*

- (I) Multiplying $x > y$ by x gives $x^2 > xy$.
- (II) Multiplying $x > y$ by y gives $xy > y^2$.
- (III) Combining, $x^2 > xy > y^2$.

Which of the following is correct?

- A The proof is valid.
- B Only line II is invalid, and the claim is true.
- C Only line III is invalid.
- D Lines I and II are invalid, but the claim is true.
- E Line I needs $x > 0$ and line II needs $y > 0$; the claim is false.
- F The proof is valid when $x, y \geq 0$, and the claim is true for all real x, y .

Question 9

A student proves that $\log_2 3$ is irrational.

- (I) Suppose $\log_2 3 = \frac{p}{q}$, where p and q are integers and $q \neq 0$.
- (II) Since $\log_2 3 > 0$, we may take p and q to be positive.
- (III) Then $2^{p/q} = 3$, so $2^p = 3^q$.
- (IV) But 2^p is even and 3^q is odd, a contradiction.

If line II were deleted, which line would no longer be justified?

- A I
- B III
- C IV
- D none: line II is unnecessary
- E both III and IV

Question 10

How many of the following statements about integers n are true?

- I n is even only if n^2 is even.
- II n^2 is divisible by 9 unless n is not divisible by 3.
- III n is prime only if n is odd.
- IV n is odd if $n^2 + 1$ is even.

- A 0
- B 1
- C 2
- D 3
- E 4

Question 11

Which of the following is the largest? (All angles are in radians.)

- A $\sin 1$
- B $\sin 2$
- C $\sin 3$
- D $\cos 1$
- E $\cos 2$

Question 12

In triangle ABC , which of the following conditions guarantee that the triangle is isosceles?

- I $\sin A = \sin B$
 - II $\cos A = \cos B$
 - III $\tan A = \tan B$
 - IV $\sin 2A = \sin 2B$
- A I only
 - B II only
 - C I and II only
 - D I, II and III only
 - E I, II, III and IV
 - F II and III only

Question 13

An odd integer N with $1 \leq N \leq 50$ is the subject of four statements:

- (1) N is divisible by 4. (2) N is divisible by 6. (3) N is a perfect square. (4) $N > 40$.

Exactly two of the four statements are true. Find N .

- A 9
- B 25
- C 36
- D 45
- E 49
- F there is no such N

Question 14

The sequence $x_1, x_2, \dots$ is defined by $x_1 = a$ and $x_{n+1} = x_n^2 - 2$. For how many of the values $a = 2, 0, 3, -2.5, 1, -2$ is the sequence bounded?

- A 1
- B 2
- C 3
- D 4
- E 5
- F 6

Question 15

The polynomial $p(x)$ has integer coefficients, and $p(0)$ and $p(1)$ are both odd. Which of the following must be true?

- I p has no integer roots.
- II p has no rational roots.
- III $p(2)$ is odd.
- IV $p(x) \neq 0$ for every real x .
- A I and III only
- B I only
- C I, II and III only
- D III only
- E I, III and IV only
- F I, II, III and IV

Question 16

The function f is continuous and satisfies $f(x) + f(1 - x) = 1$ for every real x . Which of the following must be true?

- I $\int_0^1 f(x) dx = \frac{1}{2}$
- II $f\left(\frac{1}{2}\right) = \frac{1}{2}$
- III f is increasing.
- IV $\int_0^1 xf(x) dx = \int_0^1 (1 - x)(1 - f(x)) dx$
- A I and II only
- B II only
- C I, II, III and IV
- D I and IV only
- E II and IV only
- F I, II and IV only

Question 17

Which of the following statements are true?

- I A parallelogram whose vertices lie on a circle is a rectangle.
- II A rhombus whose vertices lie on a circle is a square.
- III A trapezium whose vertices lie on a circle is isosceles.
- IV A kite whose vertices lie on a circle has two right angles.
- V A quadrilateral whose vertices lie on a circle and whose diagonals are perpendicular is a kite.
- A I, II, III and IV only
- B I and II only
- C I, II and III only
- D I, II, III, IV and V
- E I, II and IV only
- F III and V only

Question 18

Three arguments are offered for the claim *no positive integer n makes $n^2 + n + 1$ divisible by 5*.

- I.** For $n = 1, 2, 3, 4, 5$ the values are 3, 7, 13, 21, 31, none of which is divisible by 5.
- II.** Every n is congruent to 0, 1, 2, 3 or 4 modulo 5, and for these $n^2 + n + 1 \equiv 1, 3, 2, 3, 1 \pmod{5}$ respectively.
- III.** $n^2 + n + 1 = n(n + 1) + 1$, and the product of two consecutive integers ends in 0, 2 or 6, so $n^2 + n + 1$ ends in 1, 3 or 7.

Which of these are valid proofs of the claim?

- A** I only
- B** II only
- C** I and II only
- D** I, II and III
- E** II and III only
- F** none of them

Question 19

An arithmetic progression $u_1, u_2, \dots$ has non-zero common difference, and its partial sums satisfy $S_{10} = S_{20}$. Which of the following must be true?

- I** $S_{30} = 0$
 - II** $u_{15} + u_{16} = 0$
 - III** $u_{15} = 0$
 - IV** S_{15} is either the largest or the smallest of all the S_n .
- A** I only
 - B** I and II only
 - C** II and IV only
 - D** I, II, III and IV
 - E** I and III only
 - F** I, II and IV only

Question 20

Real numbers a and b satisfy $0 < a < b < 1$. Which of the following must be true?

- I** $a^b < b^a$
 - II** $a^a < b^b$
 - III** $\log_b a > 1$
 - IV** $a^b < a^a$
- A** I and III only
 - B** III and IV only
 - C** I, III and IV only
 - D** I, II, III and IV
 - E** I and IV only
 - F** II, III and IV only

END OF PAPER